Set of sets / Family of sets

$P(S)$ of power set of $S(\neq \phi)$ which is a collection of all subsets of S.

Ex: $S = \{1, 2\}$ $\quad P(S) = \{\{1\}, \{2\}, \{1, 2\}, \phi\}$

$$P(S) = \{A : A \subseteq S\}$$

Let I be a finite set $\{1, 2, \ldots, K\} =: \mathbb{N}_K$

$\mathcal{F}$ be a family of K sets $A_1, A_2, \ldots, A_K$,

$$\mathcal{F} = \{A_\alpha : \alpha \in I\}$$

Index set

(a) $I = \mathbb{N}$ $\mathcal{F}$ is an infinite collection of sets

~~(b)~~ Let I be index set and $\{A_\alpha : \alpha \in I\}$ be a family of subsets of a universal set S. Then

$$\bigcup_{\alpha \in I} A_\alpha = \{x \in S : x \in A_\alpha \text{ for at least one } \alpha \in I\}$$

$$\bigcap_{\alpha \in I} A_\alpha = \{x \in S : x \in A_\alpha \ \forall \alpha \in I\}$$

Let $\mathcal{F} = \{A_1, \ldots, A_n\}$

(a) $\bigcup_{i=1}^{n} A_i = \{x : x \in A_i \text{ for at least one } i\}$ (3)

(b) $\bigcap_{i=1}^{n} A_i = \{x : x \in A_i \ \forall i\}$

(c) $\left(\bigcup_{i=1}^{n} A_i\right)' = \bigcap_{i=1}^{n} A_i'$

(d) $\left(\bigcap_{i=1}^{n} A_i\right)' = \bigcup_{i=1}^{n} A_i'$

<u>Ex</u>: (1) Let $A_n = [0, 1/n]$ for each $n \in \mathbb{N}$

$$A_1 \supset A_2 \supset A_3 \cdots \supset A_n \supset \cdots$$

$$\bigcap_{n \in \mathbb{N}} A_n = \{0\}$$

Ex ② Let $A_n = (0, 1/n)$

$$A_1 \supset A_2 \supset \cdots \quad \text{and} \quad \bigcap_{n \in \mathbb{N}} A_n = \phi$$

Ex-③ Let $A_n = [0, 1 - \frac{1}{n}]$ $n \in \mathbb{Z}^+$

(a) $\bigcap_{n=1}^{\infty} A_n = \{0\}$

(b) $\bigcup_{n=1}^{\infty} A_n = [0, 1)$

(4)

(c) $\bigcap_{n=1}^{\infty} A_n' = [0, 1)' = (-\infty, 0) \cup [1, \infty)$

(d) $\bigcup_{n=1}^{\infty} A_n' = \{0\}' = \mathbb{R} - \{0\}$

(4) Let $A_n = \left(2 - \frac{1}{n}, 5 + \frac{1}{n}\right)$, $n \in \mathbb{Z}^+$

(a) $\bigcap_{n=1}^{\infty} A_n =$

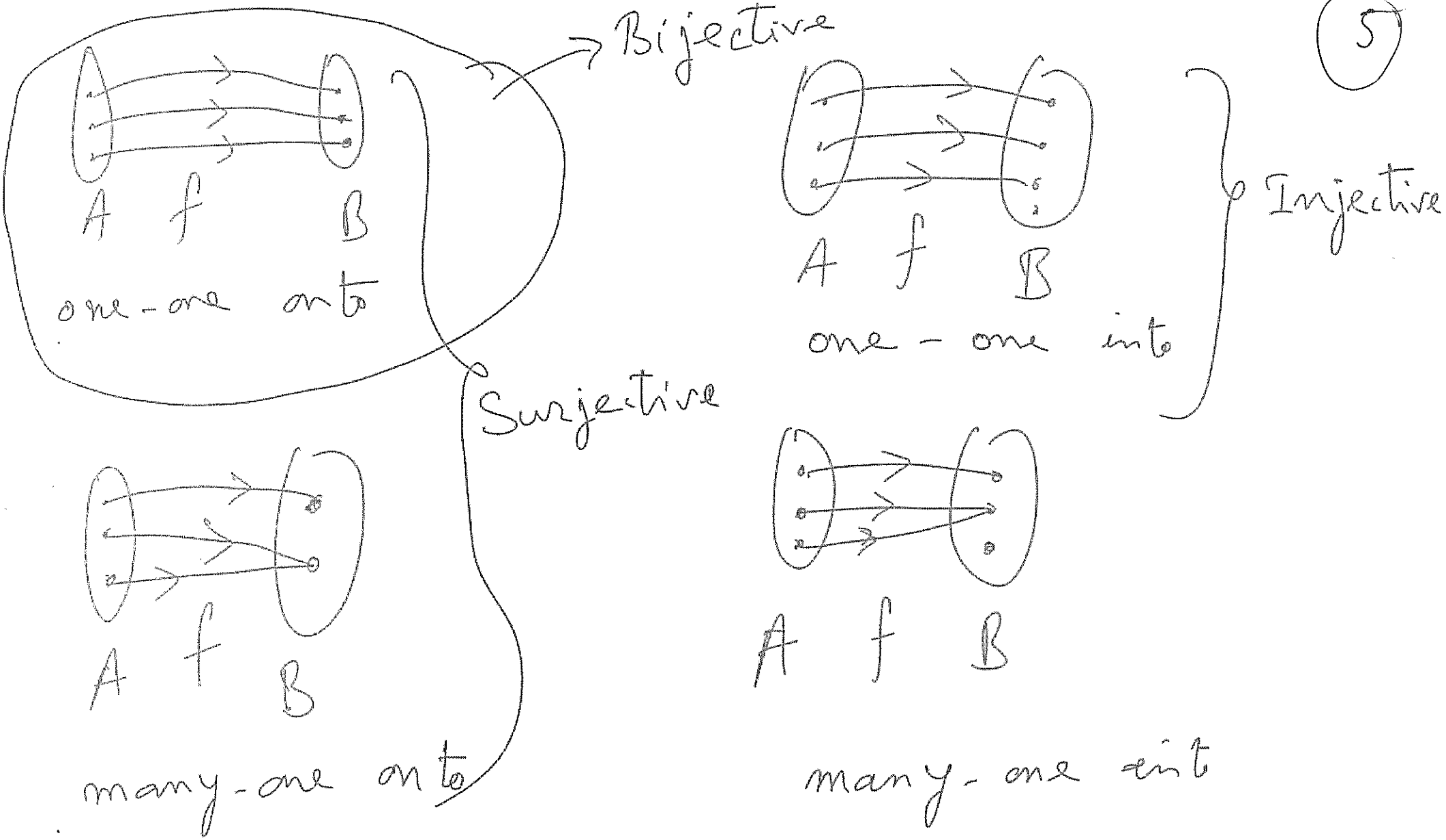
(b) $\bigvee_{n=1}^{\infty} A_n =$

(c) $\bigcap_{n=1}^{\infty} A'_n =$

(d) $\bigvee_{n=1}^{\infty} A'_n =$

Higher Algebra - Abstract and Linear - S. K. Mapa.

Naive Set Theory - Paul R. Halmos


Bijective
A f B
one-one onto
Injective
A f B
one - one into
Surjective
A f B
many-one onto
A f B
many-one into

Bijective

- Injective: $x, y \in A$ and $f(x) = f(y) \Rightarrow x = y$
- Surjective: $\forall y \in B \Rightarrow$ pre image of $y \in A$

Equipotent Set

Def: Two non empty sets A, B are said to be equipotent, if there exists a bijection $f : A \to B$.

$A \sim B$

Ex (1) $S = \{1, 1/2, 1/3, \cdots\}$, $S \sim \mathbb{N}$

Sol^{n}: $f : \mathbb{N} \to S$, $f = \frac{1}{n}$ $\forall n \in \mathbb{N}$

Let A^B be the set of all functions from B to A.

$\{0, 1\}^S \equiv$ Set of all f^{n} from S to $\{0, 1\}$

Ex (4) $P(S) \sim \{0, 1\}^S$

Soln: $F : P(S) \to \{0,1\}^S$

$\forall A \in P(S)$

$F(A) = 1_A$ ~~(x)~~

~~$x \in S$~~

$f : S \to \{0,1\}$

$f(x) = \begin{cases} 0 \\ + \end{cases}$

where $1_A(x)$ is an indicator f^{n} i.e.

$$1_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

Note $A \subseteq S$ and $x \in S$, ie x may or may not $\in A$

For $A, B \in P(S)$, if $F(A) = F(B)$

$\Rightarrow 1_A = 1_B \Rightarrow A = B$

$\Rightarrow F$ is injective (7)

Let's take a f^n: $f \in \{0,1\}^S$

$S = \{x_1, x_2, \ldots, x_i, \ldots\}$
$\exists\, i \ni f(x_i) = 1$
$f(x_i) = 0$

Now ~~Then~~ f can be express as

$$f = 1_X \quad \text{where} \quad X = \{x \in S : f(x) = 1\} \in P(S)$$

$\therefore$ For any $f \in \{0,1\}^S$ $\exists$ a pre image (in the above case X) of f under F in $P(S)$

$\Rightarrow F$ is surjective

Ex 3

$$\underset{A}{[0,1]} \sim \underset{B}{[0,3]}$$

Sol:

$$f : A \to B$$

$$f(x) = 3x, \quad x \in A$$

Ex 4

$$[0,1] \sim [a,b]$$

$$f : A \to B$$

$$f(x) = a + (b-a)x \quad \forall x \in A$$

Let $\mathbb{N}_k = \{1, 2, \ldots, k\}$

(I) A is finite if $A \sim \mathbb{N}_k$ for some $k \in \mathbb{N}$

(II) A is infinite if A is not finite.

(III) A is (d)enumerable / countably infinite if $A \sim \mathbb{N}$

Ex- $\mathbb{Z}$ is countably infinite

Sol. $f : \mathbb{N} \to \mathbb{Z}$

$$f(x) = \begin{cases} x/2 & \text{if } x \text{ is even} \\ \dfrac{1-x}{2} & \text{if } x \text{ is odd} \end{cases}$$

(IV) A is countable if A is finite or countably infinite.

(v) A is uncountable if A is not countable

$\Rightarrow$ ~~A is countable~~

$\mathbb{N} \sim A$

$f : \mathbb{N} \to A$

For each $n \in \mathbb{N}$ $\exists$ a unique $f(n) \in A$

$A = \{ f(1), f(2), \ldots, f(n), \ldots \}$

$= \{ a_1, a_2, \ldots \}$

$\{a_i\}_{i=1}$

EX: Let A is a set with atleast two elements. Then $S = A^{\mathbb{N}}$, set of all fn.s $\mathbb{N}$ into A is uncountable.

Soln. By contradiction.

Assumption: S is countably infinite.

we express S as

$$S = \{f_1, f_2, f_3, \ldots\}$$

Note: $f_i : \mathbb{N} \to A$

~~Now we construct a fn. $g: \mathbb{N} \to A$ find $g \neq f_i$ for any i and $g: \mathbb{N} \to A$ Then we are done.~~

[Now we try to find a fn. $g : \mathbb{N} \to A$ but $g \neq f_i$ for any i.
Then we are done. Aim: $g(1) \neq f_1(1)$, $g(2) \neq f_2(2)$ ... $g(i) = f_i(i)$
$\Rightarrow g \neq f_1$, $\Rightarrow g \neq f_2$ ... $\Rightarrow g \neq f_i$]

Let $a, b\ (\neq a) \in A$ and define

$\underline{g : \mathbb{N} \to A}$ by

$$\forall n \in \mathbb{N}\ g(n) = \begin{cases} b & \text{if } f_n(n) = a \\ a & \text{if } f_n(n) \neq a \end{cases}$$

~~$\forall n \in \mathbb{N}$~~

Note: This is possible since $\exists$ atleast two elements in A.

$g \neq f_n$ for each $n \in \mathbb{N}$

$\Rightarrow g \notin S$

This contradicts the fact that S is the set of all fns from $\mathbb{N}$ into A.

$\square$

Subsets of Infinite Sets

Th: Every infinite set contains a countable infinite subset.

Hints: S infinite set.

$$a_1 \in S$$

$$a_2 \in S \setminus \{a_1\}$$

$$\vdots$$

$$\{a_1, a_2, a_3, \ldots\} \subset S$$

Ex – $\mathbb{N} \subset \mathbb{Z}$

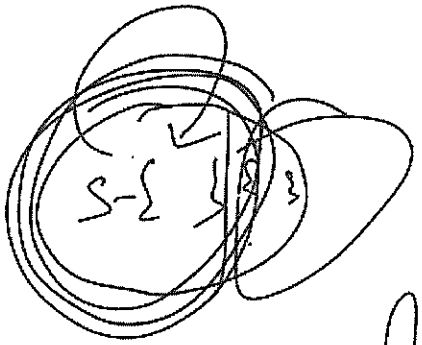
$\mathbb{N} \subset \mathbb{R}$

Th: ~~A~~ S is infinite iff it is (equipotent) equivalent to a proper subset of itself.

Hints: ($\Rightarrow$) S

$\underbrace{\{a_1, a_2, a_3, \dots\}}_{S_1 \cup S_2} \subset S$

countable infinite



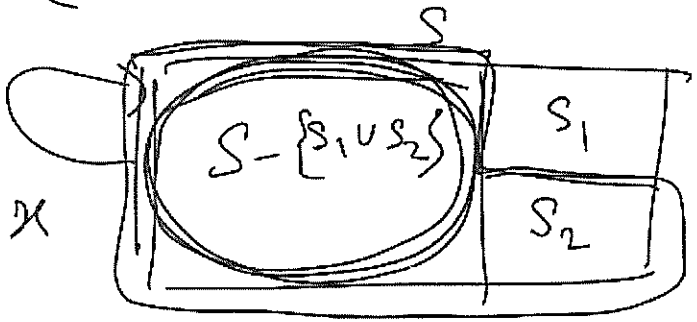
$$S \setminus \{a_1, a_3, a_5, \dots\} \subset S$$

$$f : S \to S \setminus \{a_1, a_3, a_5, \dots\}$$

$$f(x) = x, \quad \forall x \in S \setminus \{a_1, a_2, a_3, \dots\}$$

$$f(a_n) = a_{2n} \qquad a_n \in \{a_1, a_2, a_3, \dots\}$$

$f(x) = x$



$$S_1 = \{a_1, a_3, a_5 \dots\}$$
$$S_2 = \{a_2, a_4, a_6, \dots\}$$

$$S_1 \cup S_2 \longmapsto S_2$$

$$f(a_n) = a_{2n}$$

($\Leftarrow$) Prove: S is equipotent to a proper subset to itself $\Rightarrow$ S is infinite.

i.e. (contrapositive): S is finite $\Rightarrow$ S is not equipotent to any proper subset to itself, which is true by Pigeonhole principal.

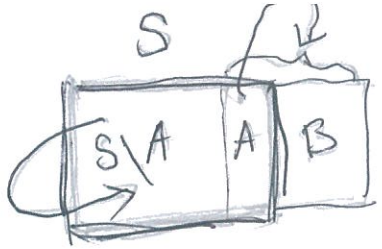
<u>Th</u>: If S is an infinite set and B is a countable set, then

$$S \sim S \cup B$$

I do it when B is inf. set. if

<u>Hint</u>: $B = \{ b_1, b_2, \ldots \}$

$$\{ a_1, a_2, \ldots \} = A \subset S$$



$$f : S \to S \cup B$$

$$f(x) = x, \ \forall x \in S \setminus \{a_1, a_2, \dots\}$$

$S \setminus A$

HW: Show $\mathbb{R} \sim \mathbb{R} \cup \{i, 2i, 3i\}$

$$f(a_n) = \begin{cases} b_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \\ a_{n/2} & \text{if } n \text{ is even} \end{cases}$$

HW: Show $\mathbb{R} \sim \mathbb{R} \setminus \{0\}$

Corr: $S \to$ uncountable

$B', B \to$ countable, $B \subseteq S$

HW

(i) $S \cup B'$, $S \setminus B$ uncountable

(ii) $S \cup B' \sim S \setminus B$?

Th: If S is countably infinite and $A \subseteq S$ then A is countable.

$A \subseteq S$, S is countable $\Rightarrow$ A is countable.

Th: Suppose $A \subseteq B$

(i) B is finite $\Rightarrow$ A is finite

(ii) A is infinite $\Rightarrow$ B is infinite

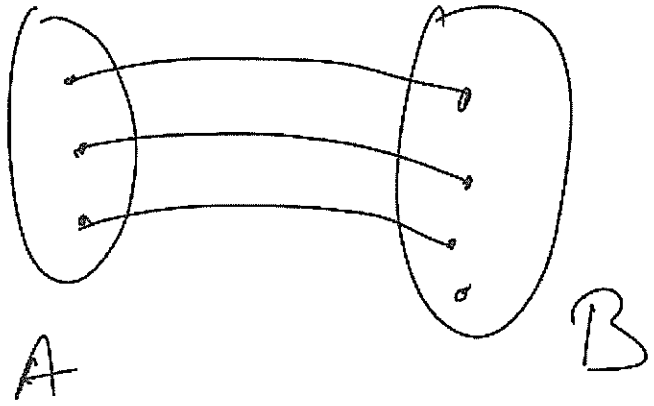
(iii) B is countable $\Rightarrow$ A is countable

(iv) A is uncountable $\Rightarrow$ B is uncountable

Th

(Injective)

one – one fn f



$f : A \to B$ ~~$\Rightarrow A = B$~~

(i) B finite $\Rightarrow$ A is finite

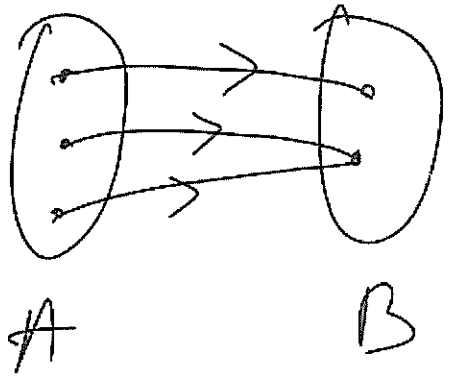
(ii) A infinite $\Rightarrow$ B is infinite

(iii) B is countable $\Rightarrow$ A is countable

(iv) A is uncountable $\Rightarrow$ B is uncountable

Th

(Surjective) <u>onto</u> func. f



$f: A \to B$ ~~($B \subseteq A$)~~

(i) A is finite $\Rightarrow$ B is finite

(ii) B is infinite $\Rightarrow$ A is infinite

(iii) A is countable $\Rightarrow$ B is countable

(iv) B is uncountable $\Rightarrow$ A is uncountable